Answers to the questions in `FPE_ch5.tex`

Presenter notes for the Chapter 5 talk

October 28, 2026

One section per `% Question` comment in the slides, in slide order, plus the questions inside some `% TODO` comments and a section on symmetries. Notation is the deck's: $\partial_t p = -\partial_x(Ap) + \frac{1}{2}\partial_x^2(Bp)$, $B = D$ for constant noise (so $\langle \Delta x^2 \rangle = Dt$), $\mathcal{U}$ the physical potential and $U = \mathcal{U}/\gamma$. Equation numbers marked G2 are Gardiner's 2nd edition.

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1 Itô, Stratonovich, and manifolds

Slide: The Corresponding SDE.

Where the two come from. An SDE is shorthand for an integral, and $\int b(x)\,\mathrm{d}W$ needs a rule for where in each step b is evaluated. Itô evaluates it at the start of the step. Then b is independent of the increment, so $\langle b\,\mathrm{d}W\rangle = 0$. Stratonovich evaluates it at the midpoint. Expanding $b(x_{\mathrm{mid}}) \approx b(x) + \frac{1}{2}b'(x)\,\Delta x$ and using $\Delta x \approx b\,\Delta W$ gives

$$b(x_{\mathrm{mid}})\,\Delta W \approx b\,\Delta W + \tfrac{1}{2}\,b\,b'\,(\Delta W)^2 \to b\,\mathrm{d}W + \tfrac{1}{2}\,b\,b'\,\mathrm{d}t.$$

So the Stratonovich SDE $\mathrm{d}x = a\,\mathrm{d}t + b \circ \mathrm{d}W$ is the Itô SDE with drift $a + \frac{1}{2}bb'$. With $b = \sqrt{B}$, $\frac{1}{2}bb' = \frac{1}{4}B'$, which is the slide's $A = a + \frac{1}{4}\partial_x B$. The FPE is always written with the Itô drift. In Stratonovich variables it has a neat "sum of squares" form:

$$\partial_t p = -\partial_x(a\,p) + \tfrac{1}{2}\,\partial_x\big[b\,\partial_x(b\,p)\big].$$

(Expand the last term to check that it equals $-\partial_x(\frac{1}{2}bb'p) + \frac{1}{2}\partial_x^2(b^2p)$.)

Which one is physical. It depends on where the noise comes from.

- Noise that is really smooth, with a short but nonzero correlation time, gives Stratonovich in the white-noise limit (the Wong–Zakai theorem). Ordinary calculus holds for smooth noise, and Stratonovich is the calculus that keeps the ordinary chain rule.
- Noise that acts in discrete kicks, with the state fixed before each kick, gives Itô. Population models and finance are like this.
- A thermal particle whose friction depends on position, $\gamma(x)$, is neither. The overdamped limit of the Langevin equation must reproduce the Boltzmann density, and that requires the Itô drift $A = -\mathcal{U}'/\gamma + \frac{1}{2}B'$, with $B = 2k_BT/\gamma(x)$. This is sometimes called the isothermal or Hänggi–Klimontovich convention. The interface section of the notes compares all three ($\chi = 0$, $\frac{1}{2}$, 1).

When B is constant all three agree, which is every example in the talk.

Manifolds: yes, this is the connection. Under a change of variable $y = \varphi(x)$, Itô's lemma gives

$$A_y = \varphi' A + \tfrac{1}{2}\varphi'' B, \qquad B_y = \varphi'^2 B.$$

B transforms as a tensor (two upper indices), but the Itô drift picks up a term with a second derivative of the coordinate change, so A^i is *not* a vector field. The Stratonovich chain rule has no

such term: $\mathrm{d}y = \varphi'(x) \circ \mathrm{d}x$, so a and b both transform like components of vector fields. That is why SDEs on a manifold are written in Stratonovich form,

$$\mathrm{d}X = V_0(X)\,\mathrm{d}t + \sum_\alpha V_\alpha(X) \circ \mathrm{d}W^\alpha,$$

with $V_0, V_1, \ldots$ vector fields. The backward operator (the generator) is then coordinate-free, $\mathcal{L}^\dagger = V_0 + \frac{1}{2}\sum_\alpha V_\alpha V_\alpha$, a sum of squares of first-order operators (Hörmander's form). Its adjoint with respect to a volume form μ is the forward operator, $\mathcal{L}\rho = -\operatorname{div}_\mu(V_0\rho) + \frac{1}{2}\sum_\alpha \operatorname{div}_\mu(V_\alpha \operatorname{div}_\mu(V_\alpha \rho))$.

Example: Brownian motion on a Riemannian manifold has generator $\frac{1}{2}\Delta_g$ (Laplace–Beltrami). In coordinates, $\frac{1}{2}\Delta_g f = \frac{1}{2} g^{ij}\partial_i\partial_j f - \frac{1}{2} g^{jk}\Gamma^i_{jk}\partial_i f$, so its Itô drift is $-\frac{1}{2} g^{jk}\Gamma^i_{jk}$: a drift with no force behind it, purely from the coordinates. The radial part of free motion in d dimensions, with drift $(d-1)D/2r$, is the simplest case.

What this means for the index notation on the slides. Writing A^i and B^{ij} with upper indices is fine in Cartesian coordinates, which is all Gardiner uses. Under nonlinear coordinate changes B^{ij} is a tensor and A^i is not. But the current $J^i = A^i p - \frac{1}{2}\partial_j(B^{ij}p)$ *is* a vector density (with p a density), as it must be if its flux through a surface is physical (slide "Flux across any surface"). In one dimension, with $p_y = p/\varphi'$, a short calculation gives $J_y = J_x$: the extra drift term cancels the extra derivative term.

2 Why the overdamped limit is $m\,\mathrm{d}v \approx 0$

Slide: The overdamped limit.

Write the velocity equation as relaxation toward the local terminal velocity $v_*(x) = -\mathcal{U}'(x)/\gamma$:

$$\mathrm{d}v = -\frac{1}{\tau_v}\big(v - v_*(x)\big)\,\mathrm{d}t + \frac{\sqrt{2\gamma k_B T}}{m}\,\mathrm{d}W, \qquad \tau_v = \frac{m}{\gamma}.$$

Integrate the momentum equation over a step Δt with $\tau_v \ll \Delta t \ll \tau_x$:

$$m\,\Delta v = -\gamma\,\Delta x - \mathcal{U}'\,\Delta t + \sqrt{2\gamma k_B T}\,\Delta W.$$

- The left side is bounded: v is a thermal velocity, so $m\,\Delta v \sim \sqrt{m k_B T}$ whatever Δt is.
- The noise impulse grows like $\sqrt{2\gamma k_B T\,\Delta t}$, and the force and friction impulses grow like Δt.
- Their ratio is $m\,\Delta v/\text{noise} \sim \sqrt{\tau_v/\Delta t} \to 0$.

So on any time step much longer than τ_v, the inertial term is negligible next to the others, and the remaining terms must balance: $\gamma\,\mathrm{d}x = -\mathcal{U}'\,\mathrm{d}t + \sqrt{2\gamma k_B T}\,\mathrm{d}W$. "$m\,\mathrm{d}v \approx 0$" does *not* mean that v is constant or small. v fluctuates wildly, and in the limit it is white noise; what is small is the momentum change compared with the impulses over a coarse step.

The upper limit $\Delta t \ll \tau_x$ is needed so that $\mathcal{U}'$ is roughly constant over the step; the window $\tau_v \ll \Delta t \ll \tau_x$ exists only if $\varepsilon = \tau_v/\tau_x = m|\mathcal{U}''|/\gamma^2 \ll 1$.

A check you can quote. For a free particle ($\mathcal{U} = 0$) the Langevin equation is solvable (Ornstein, 1917):

$$\langle \Delta x^2 \rangle = \frac{2k_B T}{\gamma}\Big[t - \tau_v\big(1 - e^{-t/\tau_v}\big)\Big] \;\to\; \begin{cases} (k_B T/m)\,t^2 & t \ll \tau_v \text{ (ballistic)} \\ (2k_B T/\gamma)\,t = Dt & t \gg \tau_v \text{ (diffusive)} \end{cases}$$

The FPE describes only the second regime; that is the "valid for $t \gg \tau_v$" on the slide.

A caution. With constant γ the naive step is safe. If γ depends on x, simply dropping $m\,\mathrm{d}v$ and reading the result as Itô gives the wrong stationary state; the correct limit has the extra drift $\frac{1}{2}B'$ (section 1).

3 Smoluchowski's equation in Gardiner

Slide: Smoluchowski's equation.

Yes, in two places (2nd edition; the 4th renumbers, so use the index under "Smoluchowski equation" or "adiabatic elimination"):

- §5.1 says some authors call the FPE the Smoluchowski equation, after Smoluchowski's early use of it.
- §6.4, "Adiabatic Elimination of Fast Variables", starts from Kramers' equation (G2 §5.3.6a) and derives the Smoluchowski equation as the leading order in $1/\gamma$ (G2 (6.4.18)). It then computes the next order, the corrected Smoluchowski equation (G2 (6.4.108)), which multiplies the diffusion by $1 + m\mathcal{U}''/\gamma^2$. That is the $1 \pm \varepsilon$ on the slide.

So your "supposedly" is right: what the slide does by hand is the leading order of Gardiner's systematic expansion. The slide now says so without the question mark.

4 The Einstein relation

Slide: Smoluchowski's equation.

In physics units, with D_{phys} defined by $\partial_t p = D_{\text{phys}}\,\partial_x^2 p$ (so $\langle \Delta x^2 \rangle = 2D_{\text{phys}}t$),

$$D_{\text{phys}} = \frac{k_B T}{\gamma} = \mu\, k_B T, \qquad \mu = \frac{1}{\gamma} \text{ (mobility: velocity per unit force).}$$

It ties the size of the fluctuations (diffusion) to the dissipation (friction): the simplest fluctuation–dissipation relation. One-line derivation: in a potential, zero current $J = -(\mathcal{U}'/\gamma)p - D_{\text{phys}}\,p' = 0$ gives $p \propto e^{-\mathcal{U}/\gamma D_{\text{phys}}}$, and this is the Boltzmann factor $e^{-\mathcal{U}/k_B T}$ only if $\gamma D_{\text{phys}} = k_B T$.

History: Einstein (1905), Sutherland (1905) and Smoluchowski (1906). With Stokes friction $\gamma = 6\pi\eta a$ it becomes the Stokes–Einstein relation $D_{\text{phys}} = k_B T/6\pi\eta a$, which Perrin used to measure Avogadro's number. In the deck's convention $B = D = 2D_{\text{phys}}$, hence "$D/2 = k_B T/\gamma$".

5 Why the continuity equation fixes J only up to a divergence-free field

Slide: Flux across any surface.

The continuity equation $\partial_t p + \partial_i J^i = 0$ involves $\boldsymbol{J}$ only through its divergence. If $\boldsymbol{J}$ satisfies it, so does $\boldsymbol{J} + \boldsymbol{K}$ for any $\boldsymbol{K}$ with $\partial_i K^i = 0$; in three dimensions, any $\boldsymbol{K} = \nabla \times \boldsymbol{M}$. The flux of $\boldsymbol{K}$ through a *closed* surface is zero (divergence theorem), so the total probability in every region comes out the same. But the flux of $\boldsymbol{K}$ through an *open* surface, such as S_{12}, need not be zero. So the continuity equation alone cannot say how much probability crosses S_{12}. Gardiner's (5.2.12) computes that crossing directly from two-time probabilities and finds that the particular $\boldsymbol{J}$ of (5.2.8) gives it. That is the sense in which it is "the physical" current.

The one-dimensional version is the clearest. Divergence-free means constant, so the continuity equation fixes J only up to an added constant. On an interval the walls fix the constant. On a circle they cannot: the free circle and the circle with constant drift v both have the uniform stationary density $p_s = 1/L$, so p_s alone cannot tell you whether probability circulates. The formula $J = vp - \frac{1}{2}Dp'$ can: $J_s = v/L$.

6 The walls figure: axis labels and $L = 2x_0$

Slide: Reflecting, absorbing, and periodic. (A `TODO` question.)

The labels are consistent, but one fact was missing from the figure itself. Lengths are in units of the start x_0, and time in units of x_0^2/D. All three panels use the *same* Brownian path.

- (a) Hard wall: the plotted path is $|x(t)|$. Reflected Brownian motion has the same law as $|W|$, which is the image solution $g(x - x_0) + g(x + x_0)$. The shaded band $x < 0$ is the forbidden side.
- (b) Cliff: the path up to its first arrival at 0 (the cross); dotted is what it would have done next.
- (c) Ring: yes, the circumference is $L = 2x_0$, i.e. $L = 2$ in these units. The ring coordinate is $x \bmod 2x_0$, which runs over $[0, 2)$; the two horizontal lines at 0 and 2 are the same point of the ring (the cut). When the blue path reaches 2 it reappears at 0, and vice versa. Dotted grey is the lift, the unwrapped path, which is the original path again.

The slides now use a copy of the figure (made by `fpe-slides/scripts/slide_figures.py` from the notes' own plotting function) whose panel (c) title says "ring of circumference $L = 2x_0$". The notes are unchanged.

7 Winding on the circle: a quick overview

Slide: Periodic boundaries, or: the circle.

What it is. A point on the circle is $x \bmod L$. Follow a path continuously and never jump back at the cut: that gives the *lift* $\tilde{x}(t) \in \mathbb{R}$, with $\tilde{x}(t) = x(t) + L\,N(t)$, where $N(t) \in \mathbb{Z}$ is the net number of times around (anticlockwise minus clockwise). The winding is $\Delta\tilde{x}(t) = \tilde{x}(t) - \tilde{x}(0)$, the net distance travelled around the ring. Panel (c) of the walls figure shows a path and its lift.

Why it is new. The position on the circle tells you where the particle is; the winding tells you how it got there. It exists only because the circle has a loop that cannot be shrunk to a point, so no interval has it. It is the observable that measures *transport* around the ring.

How it behaves. At long times it drifts and spreads,

$$\langle \Delta\tilde{x} \rangle \simeq v_{\mathrm{w}}\, t, \qquad \operatorname{Var} \Delta\tilde{x} \simeq D_{\mathrm{eff}}\, t.$$

- The winding velocity is the stationary current times the circumference, $v_{\mathrm{w}} = LJ_s$. So $v_{\mathrm{w}} \neq 0$ exactly when $\Phi = \oint 2A/B \neq 0$, the non-equilibrium case.
- The effective diffusion constant is nonzero even in equilibrium. For $B = D$ and a periodic potential U, it is the Lifson–Jackson formula $D_{\mathrm{eff}} = D/(\langle e^{2U/D} \rangle \langle e^{-2U/D} \rangle)$, with $\langle \cdot \rangle$ the average over one period. A barrier makes it exponentially small: hopping between wells.

Examples: the free circle has $v_{\mathrm{w}} = 0$ and $D_{\mathrm{eff}} = D$ (the ring changes where the particle is, not how far it goes); constant drift has $v_{\mathrm{w}} = v$; the cosine well has $D_{\mathrm{eff}} = D/I_0(\kappa)^2$.

How it is computed. Two equivalent ways.

- First passage on the cover: the time T_{w} to first get a full circumference away, in either direction, is the exit time from $(x_0 - L, x_0 + L)$ on the line, an absorbing-interval problem. Without drift, windings are a random walk of steps $\pm L$, and $D_{\mathrm{eff}} = L^2/T_{\mathrm{w}}$.
- One eigenvalue: $\langle e^{s\Delta\tilde{x}} \rangle \sim e^{\Lambda(s)t}$, where $\Lambda(s)$ is the top eigenvalue of the forward operator with $\partial_x \to \partial_x - s$, acting on periodic functions. Then $v_{\mathrm{w}} = \Lambda'(0)$ and $D_{\mathrm{eff}} = \Lambda''(0)$. For constant drift, $\Lambda(s) = vs + \frac{1}{2}Ds^2$.

The second way is Bloch's theorem in disguise: $\Lambda(s)$ is the lowest band of the periodic problem on the line, continued to imaginary quasi-momentum $q = -is$. D_{eff} is the curvature of the band at $q = 0$, the analogue of an inverse effective mass.

Where it matters physically. The dc voltage of a Josephson junction (the phase winds), a colloid driven around a ring trap, rotary molecular motors, phase slips of an oscillator. For an interdisciplinary audience, the one-liner is: *a ring has a hole, and probability can circulate around it; the winding counts the circulation.*

8 The squared-radius figure, panel by panel

Slide: Feller's Classification, example.

The process. Take an overdamped particle in an isotropic harmonic trap in d dimensions: each component obeys $\mathrm{d}x_i = -kx_i\,\mathrm{d}t + \sqrt{D}\,\mathrm{d}W_i$. Its squared distance from the centre, $y = \sum_i x_i^2$, obeys (Itô's lemma, $\mathrm{d}x_i^2 = D\,\mathrm{d}t$)

$$\mathrm{d}y = (dD - 2ky)\,\mathrm{d}t + 2\sqrt{D}\sum_i x_i\,\mathrm{d}W_i = (dD - 2ky)\,\mathrm{d}t + \sqrt{4Dy}\,\mathrm{d}W.$$

The last equation makes sense for any real d, which is how it also describes a population with demographic noise (d is an immigration rate; $d = 0$ is Feller's branching diffusion) and the CIR interest-rate model. Units: $\ell^2 = D/k$ for y, and $1/k$ for time. The noise $B = 4Dy$ vanishes at $y = 0$, so Feller's test decides what happens there: exit for $d \le 0$, regular for $0 < d < 2$, entrance for $d \ge 2$.

(a) Sample paths, all from $y_0 = \ell^2$.

- Blue, $d = 3$: $|\boldsymbol{x}|^2$ of a 3D trapped particle. It never reaches $y = 0$; a Brownian particle in two or more dimensions does not hit a given point. That is an entrance: unreachable from inside, but a particle started exactly at 0 would leave at once.
- Green, $d = 1$: x^2 of a 1D trapped particle. The black cross marks the first time it reaches $y = 0$ (the moment x changes sign). It then leaves again: a regular boundary, reached and left. Because $y = x^2$, this path is the *reflected* choice; the absorbed choice would stop at the cross.
- Orange, $d = 0$: Feller's branching diffusion. At $y = 0$ both drift and noise vanish, so once it arrives it stays (the thick orange line along the axis from $kt \approx 1.5$): an exit, here extinction of the population.

(b) The spectrum as a function of d. Two families of eigenvalues, in units of k:

- blue, horizontal: reflected at 0, $\lambda_n = 2nk$, including the stationary $\lambda_0 = 0$. They exist for $d > 0$, where $p_s \propto y^{d/2-1}e^{-y/\ell^2}$ is integrable at 0;
- orange, falling: absorbed at 0, $\lambda_n = (2n + 2 - d)k$. They exist for $d < 2$, where the origin can be reached. The lowest, $(2 - d)k$, is the decay rate of the survival probability.

The shading is Feller's classification read off the spectrum. In the exit window only the absorbed family exists: nothing to choose. In the entrance window only the reflected family: the origin is never reached, so there is nothing to absorb, and the absorption rate $(2-d)k$ has fallen to 0 at $d = 2$. In the regular window both exist, and choosing the boundary condition chooses the spectrum. A consistency check: at $d = 1$ the reflected rates $0, 2k, 4k, \ldots$ are the *even* OU levels and the absorbed rates $k, 3k, 5k, \ldots$ the *odd* ones. That is the reflection symmetry of section 15.

(c) The density near the boundary. $\ell^2 p(y, t \,|\, y_0)$ at $kt = 0.25$, on log–log axes, to show the power laws as $y \to 0$:

- reflected: $p \propto y^{d/2-1}$. For $d = 3$ it vanishes like $y^{1/2}$ (blue); for $d = 1$ it diverges like $y^{-1/2}$ (solid green), which is just the Jacobian of $y = x^2$: $p_y = p_x/2\sqrt{y}$ with p_x finite at $x = 0$;
- absorbed: p tends to a nonzero constant (dashed green, $d = 1$; dashed orange, $d = -1$).

The lesson: where B vanishes, "absorbing" does not mean $p = 0$, contrary to Gardiner's (5.2.14), which assumes $B > 0$. What vanishes is p/p_s, the backward function. In the x variable the absorbed density is $2\sqrt{y}\, p_y \propto |x|$, which does vanish at the origin, as an absorbing wall with $B > 0$ should.

9 Adjoints: do they need an inner product, and when are $\mathcal{L}$ and $\mathcal{L}^\dagger$ adjoint?

Slides: It's Linear Algebra Time. (A `TODO` question and a `Question`.)

No inner product is needed. The general notion is the Banach-space adjoint (or transpose). Given a space X and a bilinear pairing $\langle f, p\rangle$ between X and a second space Y (typically its dual), the adjoint of $T : X \to X$ is the operator $T^\dagger$ on Y with $\langle f, Tp\rangle = \langle T^\dagger f, p\rangle$ for all p and f. The Hilbert-space adjoint is the special case $Y = X$, with the inner product as the pairing.

Here the natural pairing is the expectation value,

$$\langle f, p\rangle = \int f(x)\, p(x)\, \mathrm{d}x = \mathbb{E}_p[f],$$

between densities, $p \in L^1$ (absolutely integrable), and bounded observables, $f \in L^\infty$. L^∞ is the dual of L^1, so this is exactly the setting of a space and its dual. Probabilists say the same thing: $\mathcal{L}^\dagger$ is the *generator*, acting on observables, and $\mathcal{L}$ is its dual, acting on measures. The dagger suggests a Hilbert space, but none is involved until we want orthogonality (section 10).

Yes: adjoint exactly when the boundary term vanishes. The integration by parts (Lagrange's identity)

$$\langle f, \mathcal{L}p\rangle - \langle \mathcal{L}^\dagger f, p\rangle = \Big[-fJ - \tfrac{1}{2}Bp\,\partial_x f\Big]_a^b$$

holds for any smooth p and f, whatever their boundary values. The differential expressions are therefore always *formal* adjoints. But an operator is a differential expression *together with its domain*: the functions it is allowed to act on, which is where the boundary conditions live. $\mathcal{L}$ (with its domain) and $\mathcal{L}^\dagger$ (with its domain) are adjoint exactly when the boundary term vanishes for every p in the first domain and every f in the second. Given the forward boundary conditions, this *determines* the backward ones; this is how Gardiner derives them in §5.2.4 (G2 (5.2.65)–(5.2.69)):

- absorbing, $p = 0$: the term is $-fJ$, and J at the wall is not zero, so we need $f = 0$;
- reflecting, $J = 0$: the term is $-\frac{1}{2}Bp\,\partial_x f$, and p at the wall is not zero, so we need $\partial_x f = 0$ (for $B \neq 0$);
- circle: the end points are the same point and everything is periodic, so the terms at a and b cancel;
- infinity: p and J decay, and the terms vanish for bounded f and f';
- singular boundaries where $B \to 0$ (Feller): the term can vanish by itself, and that is what "no condition needed" means for entrance and natural boundaries;
- a partially absorbing wall, $J = \kappa p$, pairs with a Robin condition on f, $\frac{1}{2}B\,\partial_x f = -\kappa f$.

Mismatched domains (say, forward reflecting with backward absorbing) are not adjoint, and the biorthogonality $\int P_\lambda Q_{\lambda'} = \delta_{\lambda\lambda'}$ would fail.

10 What is L^2? The three spaces

Slide: Biorthogonality.

$L^2(a,b)$ is the space of square-integrable functions, $\int f^2\,\mathrm{d}x < \infty$, with the inner product $\langle f, g\rangle = \int fg\,\mathrm{d}x$. It is a Hilbert space. "$\mathcal{L}$ is symmetric in L^2" means $\langle g, \mathcal{L}f\rangle = \langle \mathcal{L}g, f\rangle$ for all f and g in its domain. A weighted $L^2(w)$ has $\langle f, g\rangle_w = \int fg\,w\,\mathrm{d}x$. Three spaces matter:

Space	Contains	Operator	Used for
L^1	densities	$\mathcal{L}$	probability: stationary states, QSDs
L^∞	bounded observables	$\mathcal{L}^\dagger$	survival, exit times, splitting
$L^2(1/p_s)$	$\int p^2/p_s < \infty$	$\mathcal{L}$ (symmetric if $J_s = 0$)	orthogonality, gaps

- In plain $L^2(\mathrm{d}x)$, $\mathcal{L}$ is not symmetric unless p_s is uniform. Its adjoint there is $\mathcal{L}^\dagger$, the same expression as before, since both pairings are $\int fg\,\mathrm{d}x$. So Q_λ, the eigenfunctions of $\mathcal{L}^\dagger$, are the left eigenvectors of $\mathcal{L}$, and P_λ are its right eigenvectors.
- In $L^2(1/p_s)$, with zero stationary current, $\mathcal{L}$ is symmetric: $\int g\,\mathcal{L}p/p_s\,\mathrm{d}x = \int p\,\mathcal{L}g/p_s\,\mathrm{d}x$. That gives real eigenvalues, orthogonal eigenfunctions ($\int P_\lambda P_{\lambda'}/p_s = \delta_{\lambda\lambda'}$, the third form on the slide), and completeness: the spectral theorem.
- The spaces are linked by two maps. $p \mapsto p/p_s$ takes $L^2(1/p_s)$ to $L^2(p_s)$ and turns $\mathcal{L}$ into $\mathcal{L}^\dagger$ (Gardiner's $p = p_s q$, G2 (5.2.71)–(5.2.72)). $p \mapsto p/\sqrt{p_s}$ takes it to plain L^2, where $\mathcal{L}$ becomes minus a Schrödinger operator, $H = -\frac{1}{2}D\partial_x^2 + V$ with $V = A^2/2D + A'/2$.
- The weighted norm has a meaning: $\int (p - p_s)^2/p_s\,\mathrm{d}x$ is the χ^2 distance from equilibrium, and it only decreases, $\frac{\mathrm{d}}{\mathrm{d}t}\int p^2/p_s\,\mathrm{d}x = -\int Bp_s\left[\partial_x(p/p_s)\right]^2\mathrm{d}x \leq 0$. That is why all $\lambda \geq 0$.

On a bounded domain all three spaces give the same discrete spectrum. On unbounded domains they can differ; the continua of QSDs on the "caution" slide are positive eigenfunctions in L^1 at λ that are not in the L^2 spectrum.

11 The propagator line of the quantum-mechanics table

Slide: Analogy with Quantum Mechanics.

In quantum mechanics the propagator is the amplitude to go from x_0 to x in time t,

$$K(x,t;x_0) = \langle x|e^{-iHt/\hbar}|x_0\rangle = \sum_n \langle x|n\rangle\langle n|x_0\rangle\, e^{-iE_nt/\hbar} = \sum_n \phi_n(x)\,\phi_n^*(x_0)\,e^{-iE_nt/\hbar},$$

obtained by inserting the resolution of the identity $\sum_n |n\rangle\langle n| = 1$. Any wavefunction evolves by $\psi(x,t) = \int K(x,t;x_0)\,\psi(x_0,0)\,\mathrm{d}x_0$.

The FPE line is the same construction. $p(x,t\,|\,x_0,0)$ is the kernel of $e^{t\mathcal{L}}$, and the resolution of the identity is the completeness relation $\sum_\lambda P_\lambda(x)\,Q_\lambda(x_0) = \delta(x-x_0)$ (the $t=0$ case). Any density evolves by $p(x,t) = \int p(x,t\,|\,x_0,0)\,p(x_0,0)\,\mathrm{d}x_0$.

The differences are instructive.

- The "bra" is ϕ_n^* in quantum mechanics because H is Hermitian: its left eigenvectors are the conjugates of its right ones. $\mathcal{L}$ is not symmetric in plain L^2, so its left eigenvectors are a different set, the Q_λ.
- K is an amplitude; $p(x,t\,|\,x_0,0)$ is a probability density. It is positive, and $\int p\,\mathrm{d}x = 1$ because $Q_0 = 1$ and $\int P_\lambda\,\mathrm{d}x = 0$ for $\lambda \neq 0$.
- Setting $t = -i\tau$ turns K into the Euclidean (heat) kernel $\sum_n \phi_n(x)\phi_n^*(x_0)e^{-E_n\tau/\hbar}$. Through $p = \sqrt{p_s}\,\psi$ the FPE propagator is exactly that kernel for H, dressed with the ground state: $p(x,t\,|\,x_0,0) = \sqrt{p_s(x)/p_s(x_0)}\ \sum_\lambda \psi_\lambda(x)\psi_\lambda(x_0)e^{-\lambda t}$.

12 The spectra figure, column by column

Slide: Example spectra.

Each column is one problem, with its eigenvalues in its own natural unit (written under it). Black ticks are discrete eigenvalues; shaded bands are continua; the orange arrow is the gap that controls the long-time behaviour.

1. Reflecting interval: $0, 1, 4, 9, \ldots$ in units of $\pi^2 D/2L^2$. $\lambda = 0$ is the stationary state; the arrow is the relaxation rate λ_1.
2. Absorbing interval: $1, 4, 9, \ldots$ No zero: nothing survives. The lowest level is the decay rate of S, and the arrow $\lambda_2 - \lambda_1 = 3$ is the rate of approach to the QSD.
3. Free half-line: a continuum down to 0, no gap, so relaxation is algebraic ($t^{-1/2}$).
4. Drift into a reflecting floor (sedimentation): the stationary $\lambda = 0$, isolated below a continuum that starts at $v^2/2D = 1/2\tau$. The gap is the edge of the continuum.
5. OU on the line: $0, 1, 2, \ldots$ in units of k. Gap k on an infinite domain, because the trap confines.
6. Harmonic well, reflecting at the bottom and absorbing at $L = 2.5\ell$ (the "escape from a trap" example).

The last column. Right: the lowest eigenvalue is not zero. The wall at 2.5ℓ is absorbing, so there is no stationary state and no $\lambda = 0$; every particle eventually escapes. But the lowest eigenvalue is tiny, $\lambda_0 \approx 0.005\,k$, too small to see on the axis, so the grey arrow labels it. It is the Kramers escape rate: $\frac{2k}{\sqrt{\pi}}\frac{L}{\ell}e^{-L^2/\ell^2}$ with $L/\ell = 2.5$ gives about $0.005\,k$ after the first correction. The next level is at about $2.2\,k$, so the orange arrow, $\lambda_1 - \lambda_0$, is about 450 times λ_0. That ratio is the metastability:

the particle equilibrates in the well (rate $\approx 2.2k$) long before it escapes (rate $0.005k$). Compare the reflecting half-line, $0, 2k, 4k, \ldots$: a far-away cliff barely moves the upper levels but turns the 0 into an exponentially small escape rate. SDE Explorer, opened from the "escape from a trap" slide, prints $\lambda_0 = 0.004954$ and $\lambda_1 - \lambda_0 = 2.232$.

13 Why $\lambda_2 - \lambda_1$ sets the approach to the QSD

Slide: Quasi-Stationary Distributions, absorbing walls.

Expand and divide:

$$\tilde{p} = \frac{p}{S} = \frac{c_1 P_1 e^{-\lambda_1 t} + c_2 P_2 e^{-\lambda_2 t} + \cdots}{c_1 \bar{P}_1 e^{-\lambda_1 t} + c_2 \bar{P}_2 e^{-\lambda_2 t} + \cdots} = \frac{P_1}{\bar{P}_1} \cdot \frac{1 + \frac{c_2 P_2}{c_1 P_1}\, e^{-(\lambda_2 - \lambda_1)t} + \cdots}{1 + \frac{c_2 \bar{P}_2}{c_1 \bar{P}_1}\, e^{-(\lambda_2 - \lambda_1)t} + \ldots},$$

with $c_n = Q_n(x_0)$ and $\bar{P}_n = \int P_n \, \mathrm{d}x$. Dividing by S removes the common factor $e^{-\lambda_1 t}$, the overall loss of probability. What is left is how fast the second mode dies *relative to the first*, and that is $e^{-(\lambda_2 - \lambda_1)t}$. So $\tilde{p} = \mathrm{QSD} + O(e^{-(\lambda_2 - \lambda_1)t})$.

It is the same reason the power method of linear algebra converges at the ratio of the top two eigenvalues: normalising at each step discards the overall scale, and only the relative weights evolve. If the start is symmetric and $c_2 = 0$, the rate is $\lambda_3 - \lambda_1$ instead (section 15).

14 Plot Lab and SDE Explorer in the talk

Slide: QSD example. (A `TODO` question.)

Yes, the QSD example is a very good spot. In SDE Explorer you watch sample paths get killed at the walls while the histogram of survivors settles onto the QSD, and the exit rate $-\dot{S}/S$ climbs to λ_1 live. Plot Lab's "free diffusion, absorbing walls" page shows the same exact curves with sliders for x_0, L and D.

The slides now have small buttons at the top right of these frames:

- Plot Lab only: the walls figure (`physical`), the squared-radius example (`sqrt`), the spectra (`spectra`), the reductions tree (`reductions`: each box of the map links to its page), walls by images (`free-summary`);
- Plot Lab and SDE Explorer: the QSD example, the box, the ring, constant drift, OU, and escape from a trap.

Every Plot Lab page also has its own "Open in SDE Explorer" button. I checked two of the links (the absorbing interval in Plot Lab, and the escape problem in SDE Explorer) with both dev servers running; they open the right process and walls.

For the talk:

- Run `npm run dev` in `plot-lab/` and `sde-explorer/` beforehand (ports 5174 and 5173), or deploy the built apps side by side and change the two URLs in the macros `\plotlab` and `\sdeexplorer` at the top of the `.tex` file.
- Some PDF viewers ask before opening an external link; do a practice click in the viewer you will present with.

15 Symmetries in the solutions

New slides: Symmetries (three frames), and remarks in the examples.

A symmetry of the problem (domain, walls, A and B) is an operation that commutes with $\mathcal{L}$. Commuting operators can be diagonalised together, so symmetries label the eigenfunctions, forbid some overlaps (selection rules), and let one solve a big problem from a smaller one. Each one below shows up somewhere in the examples.

15.1 Reflection

- **Parity.** If the domain is symmetric about c, B is even and A is odd, the reflection $R : x \mapsto 2c - x$ commutes with $\mathcal{L}$. Eigenfunctions are even or odd about c: cos and sin in the box, H_n of definite parity for OU.
- **Selection rules.** A start at c, or any even initial density, has no overlap with odd modes; an even observable never sees them. The box started at its centre relaxes at $\lambda_2 = 4\lambda_1$; in OU, $\langle x \rangle$ is a single exponential at k (only H_1), while $\langle x^2 \rangle$ relaxes at $2k$ (only H_2).
- **Sectors are walls.** An even density has $J = 0$ at c; an odd one has $p = 0$ there. So the even sector of a symmetric problem is the half problem with a reflecting wall at c, and the odd sector is the half problem with an absorbing wall. That is the method of images (a + image or a − image), and it is "folding" in the notes: the mixed interval unfolds into an absorbing interval of twice the length; the squared-radius spectrum at $d = 1$ splits into even and odd OU levels.
- **The double well as a tunnelling doublet.** In the symmetrised (Schrödinger) form, the ground state $\sqrt{p_s}$ is even and the first excited state is odd, nearly $\pm\sqrt{p_s}$ in the two wells. In quantum mechanics this pair is the tunnelling doublet, split by $\Delta E \propto e^{-S/\hbar}$. Here the splitting is $\lambda_1 = 2r$, twice the Kramers hopping rate, and the odd backward eigenfunction $Q_1 \approx 2\pi_c - 1$ is the committor. Kramers' formula is the FPE's version of a WKB tunnelling estimate.

15.2 Translations

- **Fourier modes.** On the line with constant coefficients, translations commute with $\mathcal{L}$: eigenfunctions are e^{ikx}, and p depends only on $x - x_0$.
- **The circle as a quotient.** The circle is the line modulo translations by L, so its propagator is the line's summed over all translates (the cover), every image with a + sign. A mirror imposes a condition and can flip a sign; a translation imposes nothing, it is just another way of arriving at the same point.
- **Doublets.** With rotations and a reflection, each nonzero level of the free circle is doubly degenerate: $\cos q_m x$ and $\sin q_m x$ about any point, which are the reflecting and the absorbing intervals of length $L/2$. In the notes' cosine well the doublets survive (exactly) without full rotation symmetry; a $\cos 2x$ term would split them.
- **Bloch.** With periodic coefficients on the line, translations by L still commute with $\mathcal{L}$. Eigenfunctions are Bloch waves, and the winding statistics come from the lowest band near zero quasi-momentum (section 7): D_{eff} is its curvature.

15.3 Boosts, gauges and scaling

- **Galilean boost.** Constant drift on the line is free motion in the frame $x - vt$. It fails with walls (they would move), and on the circle it is the only way to remove the drift, since the gauge factor

is not periodic.

- **The gauge** $p = e^{\alpha(x-x_0)-t/2\tau}u$ is not a symmetry but a similarity transformation: it maps the drift problem onto the free one, shifting every rate by $1/2\tau$ and turning reflecting walls into Robin walls. It is the symmetrisation $p = \sqrt{p_s}\psi$ for constant A.
- **Diffusive scaling.** $x \to cx$, $t \to c^2 t$ leaves the free equation unchanged. So the free propagator is $(Dt)^{-1/2}$ times a function of $(x - x_0)/\sqrt{Dt}$; the half-line survival $\operatorname{erf}(x_0/\sqrt{2Dt})$ depends only on x_0^2/Dt; any problem with one length L has rates $\propto D/L^2$ and times $\propto L^2/D$. Drift and confinement break the scaling by adding their own scales, $D/|v|$ and $\sqrt{D/k}$.
- **OU's map** $y = xe^{kt}$, with the clock $\theta = (e^{2kt} - 1)/2k$, is a time-dependent rescaling that turns OU into free diffusion. It keeps a wall at $x = 0$ fixed (it is a fixed point of the rescaling) and moves any other wall, which is why the escape problem stops at Kummer functions.

15.4 Time reversal and detailed balance

- A stationary process is *reversible* if a film of it run backwards has the same statistics (detailed balance):

$$p(x, t \,|\, y, 0)\, p_s(y) = p(y, t \,|\, x, 0)\, p_s(x).$$

- In the expansion this holds exactly when $P_\lambda = p_s Q_\lambda$. Then

$$P_\lambda(x)\, Q_\lambda(y)\, p_s(y) = p_s(x)\, Q_\lambda(x)\, Q_\lambda(y)\, p_s(y),$$

  which is symmetric in $x \leftrightarrow y$. So Gardiner's relation $P = p_s Q$ *is* time-reversal symmetry. Equivalently, $J_s = 0$; equivalently, $\mathcal{L}$ is symmetric in $L^2(1/p_s)$, so the spectrum is real.
- In one dimension with a wall (or $J(\pm\infty) = 0$), J_s is a constant that vanishes at the wall, so detailed balance is automatic. Gardiner says it is "almost trivial in one variable systems".
- Breaking it needs a loop or more dimensions. On a circle the obstruction is $\Phi = \oint 2A/B$; a driven ring has $J_s \neq 0$, complex-conjugate pairs of eigenvalues ($\lambda_m = \frac{1}{2}Dq_m^2 + ivq_m$ for constant drift), and a density that circulates as it relaxes. In n dimensions the obstruction is a curl: detailed balance needs $B^{-1}(2\boldsymbol{A} - \partial \cdot B)$ to be a gradient (Gardiner's potential conditions).

15.5 Hidden symmetries

- **Forward–backward duality.** $\mathcal{L}$ and $\mathcal{L}^\dagger$ have the same spectrum, with $P_\lambda \leftrightarrow Q_\lambda$ and the wall conditions paired (section 9).
- **Partner potentials (supersymmetry).** With $B = D$, the symmetrised operator factorises, $H = \mathcal{A}^\dagger\mathcal{A}$ with $\mathcal{A} = \sqrt{D/2}\,(\partial_x - A/D)$, and $\mathcal{A}\sqrt{p_s} = 0$. The partner $\mathcal{A}\mathcal{A}^\dagger$ is the same problem with $U \to -U$, and the two share every nonzero rate. For OU the partner of the trap is the inverted trap, whose rates are $k, 2k, \ldots$: OU's ladder with the 0 removed. In the notes this explains the "ladders by algebra" box of the tree, and why $U \to -U$ swaps exit and entrance boundaries.
- **Coordinate invariance.** Feller's type of a boundary does not change under a change of variable ($y = r^2$, Lamperti, $\theta = 2\arctan x$), because his integrals do not. A boundary can look regular in one chart and singular in another, but its type belongs to the process. Stratonovich form makes the same point for the SDE itself (section 1).